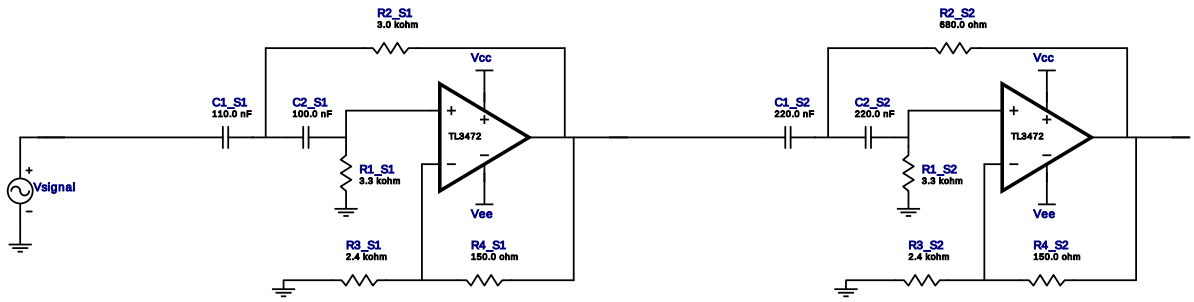
Type : Highpass
Response : Butterworth
Order : 4
Number of Stages : 2

Filter Design Report

Design : Highpass Filter - 4th order Butterworth
Design ID: 11



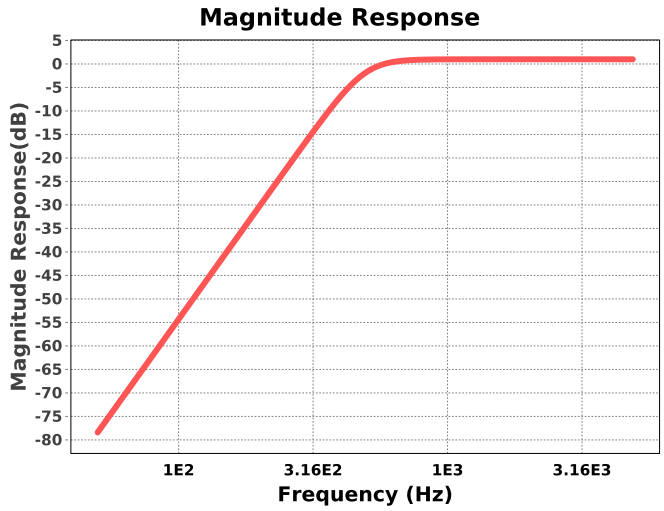
Electrical BOM

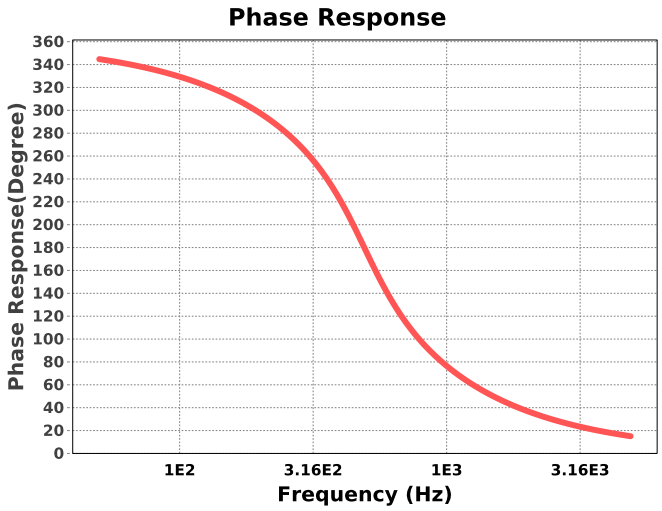
#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S1	Texas Instruments Inc.	TL3472	GbwTyp= 4MHz VccMax= 36V VccMin= 4V	1
2.	A1_S2	Texas Instruments Inc.	TL3472	GbwTyp= 4MHz VccMax= 36V VccMin= 4V	1
3.	C1_S1	Generic	Ideal	Cap= 110.0 nF Tolerance= 2.0 %	1
4.	C1_S2	Generic	Ideal	Cap= 220.0 nF Tolerance= 2.0 %	1
5.	C2_S1	Generic	Ideal	Cap= 100.0 nF Tolerance= 2.0 %	1
6.	C2_S2	Generic	Ideal	Cap= 220.0 nF Tolerance= 2.0 %	1
7.	R1_S1	Generic	Ideal	Res= 3300.0ohm Tolerance= 1%	1
8.	R1_S2	Generic	Ideal	Res= 3300.0ohm Tolerance= 1%	1
9.	R2_S1	Generic	Ideal	Res= 3000.0ohm Tolerance= 1%	1
10.	R2_S2	Generic	Ideal	Res= 680.0ohm Tolerance= 1%	1
11.	R3_S1	Generic	Ideal	Res= 2400.0ohm Tolerance= 1%	1
12.	R3_S2	Generic	Ideal	Res= 2400.0ohm Tolerance= 1%	1
13.	R4_S1	Generic	Ideal	Res= 150.0ohm Tolerance= 1%	1
14.	R4_S2	Generic	Ideal	Res= 150.0ohm Tolerance= 1%	1

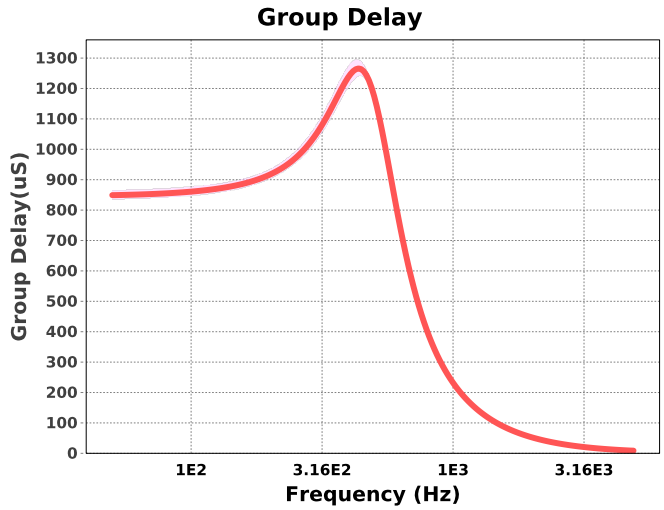
Copyright © 2026, Texas Instruments Incorporated

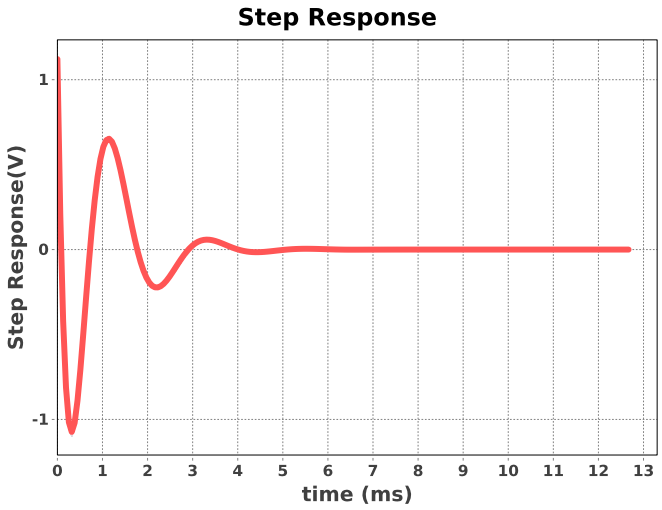
Sensitivity Analysis

#	Name	Series	Tolerance
1.	Cap	E48	2%
2.	Res	E96	1%









Copyright © 2026, Texas Instruments Incorporated

Design Inputs

#	Name	Value	Description
1.	FilterType	highpass	
2.	FilterResponse	Butterworth	
3.	FilterOrder	4.0	
4.	FilterTopology	Sallen-Key	
5.	NumberOfStages	2.0	
6.	PassbandFrequency	500.0	
7.	StopbandAttenuation	-79.999	
8.	StopbandFrequency	50.0	
9.	Gain	1.122	
10.	DualSupply	+/-5.00 V	Power supply(s) to active chips
11.	ResistorTolerance	E96	Resistor series - 1% Passive resistor tolerance
12.	CapacitorTolerance	E48	Capacitor series - 2% Passive capacitor tolerance

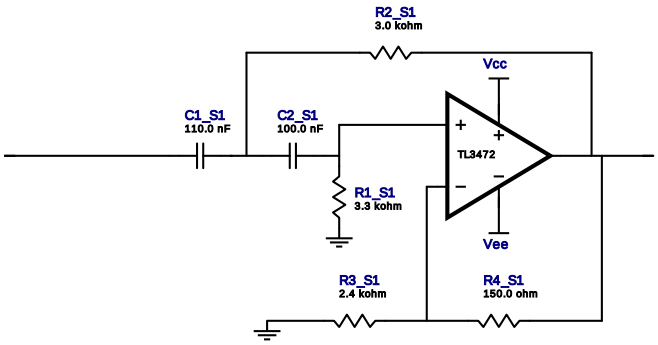
Design Assistance

1. **TL3472** Product Folder : http://www.ti.com/product/TL3472 : contains the data sheet and other resources.

Copyright © 2026, Texas Instruments Incorporated

Filter Stage :1

Cutoff Frequency	482.288 Hz
Min GBW Reqd	28.663 kHz
Stage Gain	1.062 V/V
Stage Q	541.538 m
Stage Topology	Sallen-Key



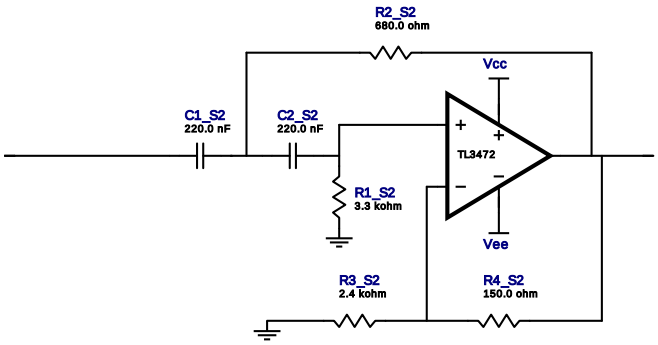
Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S1	Texas Instruments Inc.	TL3472	GbwTyp= 4MHz VccMax= 36V VccMin= 4V	1
2.	C1_S1	Generic	Ideal	Cap= 110.0 nF Tolerance= 2.0 %	1
3.	C2_S1	Generic	Ideal	Cap= 100.0 nF Tolerance= 2.0 %	1
4.	R1_S1	Generic	Ideal	Res= 3300.0ohm Tolerance= 1%	1
5.	R2_S1	Generic	Ideal	Res= 3000.0ohm Tolerance= 1%	1
6.	R3_S1	Generic	Ideal	Res= 2400.0ohm Tolerance= 1%	1
7.	R4_S1	Generic	Ideal	Res= 150.0ohm Tolerance= 1%	1

Copyright © 2026, Texas Instruments Incorporated

Filter Stage :2

Cutoff Frequency	482.932 Hz
Min GBW Reqd	69.196 kHz
Stage Gain	1.062 V/V
Stage Q	1.298
Stage Topology	Sallen-Key



Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S2	Texas Instruments Inc.	TL3472	GbwTyp= 4MHz VccMax= 36V VccMin= 4V	1
2.	C1_S2	Generic	Ideal	Cap= 220.0 nF Tolerance= 2.0 %	1
3.	C2_S2	Generic	Ideal	Cap= 220.0 nF Tolerance= 2.0 %	1
4.	R1_S2	Generic	Ideal	Res= 3300.0ohm Tolerance= 1%	1
5.	R2_S2	Generic	Ideal	Res= 680.0ohm Tolerance= 1%	1
6.	R3_S2	Generic	Ideal	Res= 2400.0ohm Tolerance= 1%	1

Copyright © 2026, Texas Instruments Incorporated

#	Name	Manufacturer	Part Number	Properties	Qty
7.	R4_S2	Generic	Ideal	Res= 150.0ohm Tolerance= 1%	1

Important Notice and Disclaimer

TI provides technical and reliability data (including datasheets), design resources (including reference designs), application or other design advice, web tools, safety information, and other resources AS IS and with all faults, and disclaims all warranties. These resources are intended for skilled developers designing with TI products. You are solely responsible for (1) selecting the appropriate TI products for your application, (2) designing, validating and testing your application, and (3) ensuring your application meets applicable standards, and any other safety, security, or other requirements.

These resources are subject to change without notice. TI grants you permission to use these resources only for development of an application that uses the TI products described in the resource. Other reproduction and display of these resources is prohibited. No license is granted to any other TI intellectual property right or to any third party intellectual property right. TI disclaims responsibility for, and you will fully indemnify TI and its representatives against, any claims, damages, costs, losses, and liabilities arising out of your use of these resources.

Providing these resources does not expand or otherwise alter TI's applicable Terms of Sale or other applicable terms available either on ti.com or provided in conjunction with TI products.

Copyright © 2026, Texas Instruments Incorporated